

Tide retrospective: each coefficient is a limit of the data

Tide loop (Claude Fable 5, with GPT-5.6 Sol deliberation)

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1 Setting

The all-order expansion tide¹ deliberated three candidates and deferred the third, the **Tendsto** packaging, until the expansion API stabilized. It has, so this tide closes it on the inherited deliberation: no fresh consult, the route was agreed in the recorded round.

2 The thing formalised

One theorem in `Laplace/OneD/RecoveryCoeffLimit.lean`², sorry-free with zero warnings: for $b \geq 0$ and every n ,

$$\left(Z_b(t) - \sqrt{2\pi} \sum_{j=0}^n \frac{(-b)^j (4j-1)!!}{j!} t^{-(j+1/2)} \right) t^{n+3/2} \longrightarrow \sqrt{2\pi} \frac{(-b)^{n+1} (4n+3)!!}{(n+1)!} \quad (t \rightarrow \infty).$$

Reading down the ladder, every coefficient of the expansion is recovered as a limit of partition-function values: the note’s “the expansion determines each coefficient” in the form a consumer of the data would actually use, and the analytic counterpart of the archimedean recovery theorems earlier in the thread.

3 Proof strategy

Exactly the deferred plan: apply the all-order expansion at order $n+1$; split the last term of the truncation with `Finset.sum_range_succ`, whose rescaling collapses to the constant coefficient (one `rpow` cancellation); the remaining order- $(n+1)$ remainder rescales to at most C/t , squeezed to zero by `squeeze_zero_norm`’ against $Ct^{-1} \rightarrow 0$; conclude by **Tendsto.add** and congruence on the eventually-positive ray.

4 Roadblocks and resolutions

One: a bare `simp` could not close the `rpow` exponent cancellation $t^{-(a)} \cdot t^a = 1$ after `rpow_add` (it normalized the exponent without recognising the sum as zero); the explicit `show ... = 0 by ring` then `rpow_zero` chain is deterministic. Second build was clean.

¹2026-08-09-13-40-tide-germbij-recovery-allorder.

²quartic_expansion_coefficient_limit.

5 What was Mathlib, what was new

Mathlib: the squeeze lemma, `tendsto_inv_atTop_zero`, filter congruence. Seabed: the all-order expansion, consumed at two adjacent orders simultaneously, which is the whole trick. New: nothing beyond the packaging; that is the point of a deferred-candidate tide.

6 Lessons

Deferring a candidate until its dependency stabilizes is cheap when the deliberation is recorded: this tide needed no new decisions, only execution. Explicit `show ... by ring` beats `simp` for `rpow` exponent arithmetic, consistently across the whole thread.

7 Follow-ups

- The potential-generic ladder $(x^{2k} + b x^{2m})$, now the only recorded follow-up of the recovery thread; the proofs are index-generic except for the moment bookkeeping.
- Tag the note's Section 7.4 dimension-one item with the all-order and coefficient-limit theorems once merged.